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One-Pot Transformation Strategy of *o*-Hydroxyphenyl Propargylamines for Modular Access to 3-Functionalized Benzofurans

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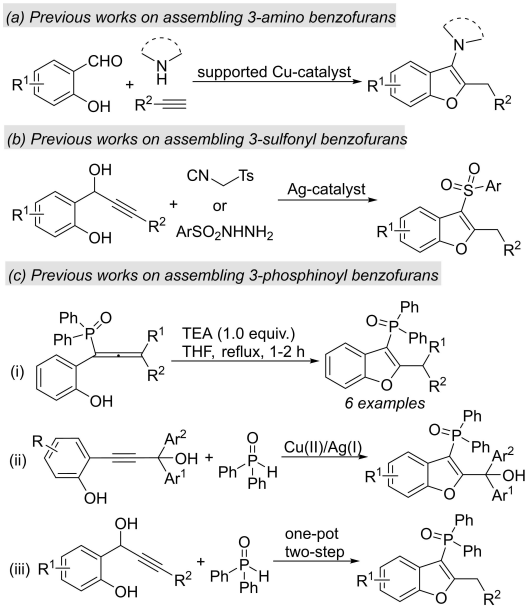
Abstract: A one-pot strategy to C3-functioanlized benzofurans is achieved from *o*-hydroxyphenyl propargylamines (*o*-HPPAs). 3-Aminobenzofurans were obtained in 64–96% yields through the cascade alkyne-allene tautomerization and 5-*exo*-*dig* annulation of *o*-HPPAs in the presence of KOH/NaO^{t}Bu as the fruitful base combination. Moreover, with benzenesulfinates and diphenylphosphine oxide as nucleophiles, a one-pot protocol towards these derivatives was also demonstrated through a cascade 1,4-conjugate addition/5-*exo*-*dig* annulation/aromatization process. Additionally, a gram-scale synthesis and further transformation into other valuable functionalized benzofurans were conducted to show the potential in organic synthesis of these developed synthetic approaches.

Keywords: Benzenesulfinates; Benzofurans; Cascade reactions; Propargylamines; Diphenylphosphine oxide

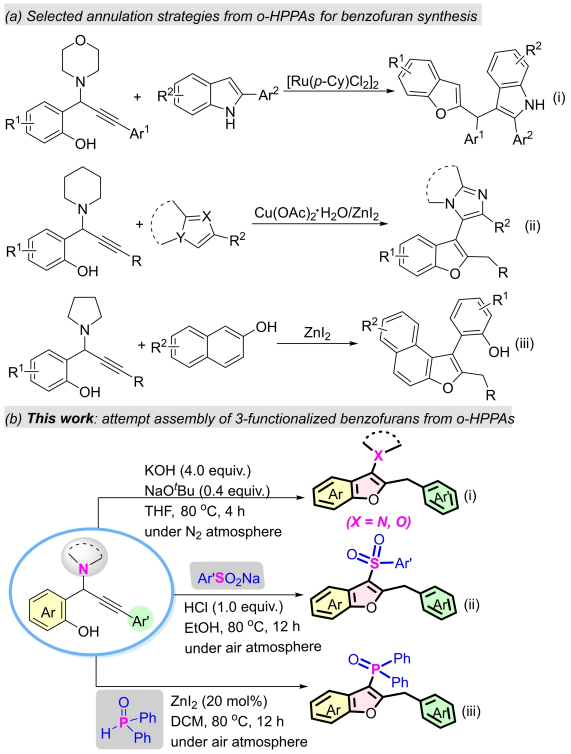
Introduction

Benzofurans are valuable and important oxygen-containing heterocycles that are found in many natural products and pharmaceuticals.[1] They exhibit a wide range of biological and pharmaceutical activities such as antifungal,[2] antitumour,[3] anti-inflammatory,[4] antiviral,[5] and antineoplastic properties.[6] Among them, C3-functionalized benzofuran derivatives have attracted considerable attention for the diverse unique properties and important bioactivities with widespread applications in drug discovery, treating diseases, the food industry, and functional materials.[7] Enjoying the discovery from brilliant chemists, modern synthetic strategies allow milder reaction conditions, yet, a more efficient way to construct versatile C3-functionalized benzofuran derivatives. Consequently, new synthetic protocols for the preparation of such compounds are being consistently reported by the chemical community.[8–10] For instance, salicylaldehydes, alkynes, and amines were subjected to a one-pot three-component reaction for the synthesis of 3-amino benzofurans in the presence of a supported copper catalyst (Scheme 1a).[11] Bi and Du, respectively, described a silver-catalyzed cascade reaction of propargylic alcohols with *p*-toluenesulfonylmethyl isocyanide or arylsulfonyl hydrazides to afford 3-sulfonyl benzofuran derivatives (Scheme 1b).[12] In 2012, Swamy developed an intramolecular cyclization of allenylphosphine oxides for assembling 3-phosphinoyl benzofurans (Scheme 1c(i)).[13] Later, an improved copper(II)-catalyzed intermolecular cascade annulation of diphenylphosphine oxide and propargylic alcohols was further investigated to generate 3-phosphinoyl benzofurans (Scheme 1c(ii)).[14] Recently, a one-pot two-step synthetic protocol for modular access to 3-

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Scheme 1. Previous works on assembling 3-functionalized benzofurans.



Scheme 2. Previous and our designed synthetic strategies using *o*-HPPAs for the synthesis of functionalized benzofurans.

phosphinoyl benzofurans was shown by Du's and Chen's group independently (Scheme 1c(iii)).[15]

Propargylamines are extremely versatile and common building blocks in the field of pharmacological and pharmaceutical chemistry.[16] Recently, *o*-hydroxyphenyl propargylamines (*o*-HPPAs) acted as *o*-AQMs (alkynyl *o*-quinone methide) precursors have been employed in numerous complex reactions for the synthesis of *oxo*-heterocyclic compounds.[17] They can react with various nucleophiles to produce structurally diverse molecules bearing furan or pyran scaffolds through the intramolecular 5-*exo*- or 6-*endo*-cyclization of the hydroxy group towards the alkynyl moiety.[18] For example, Hajra reported a Ru-catalyzed tandem furan annulation/arylation strategy to afford unsymmetrical bis(heteroaryl)methanes by using *o*-HPPAs and indoles as substrates (Scheme 2a(i)).[19] Later, a facile and efficient copper-catalyzed cascade annulation of an imidazoheterocycle with *o*-HPPAs was developed to afford biheteroaryl structural motifs containing benzofurans and imidazoheterocycles (Scheme 2a(ii)).[20] We previously employed *o*-HPPAs for the synthesis of sterically hindered heterobiaryl frameworks with benzofuran scaffold (Scheme 2a-(iii)).[21] To date, the use of *o*-hydroxyphenyl propargylamines as the base substrate for the synthesis of benzofurans functionalized by C3-hetero atoms through the direct intermolecular annulation has not been reported. Herein, we report a facile one-pot strategy to 3-aminobenzofurans through the direct intramolecular annulation of *o*-hydroxyphenyl propargylamines (Scheme 2a(i)). Moreover, with bench-top benzenesulfinates (Scheme 2b(ii)) and diphenylphosphine oxide (Scheme 2c(iii)) as nucleophiles, a one-pot protocol toward valuable 3-sulfonyl and 3-phosphinoyl benzofuran derivatives was also generated in good to high yields in the presence of hydrochloric acid and ZnI_2, respectively.

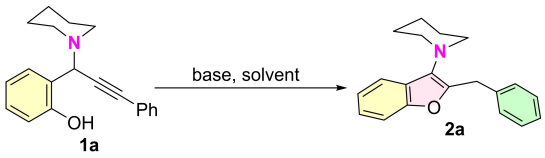
Results and Discussion

With the above-mentioned consideration, we commenced optimizing the intramolecular annulation of propargylamine **1a** as model substrate to form 3-aminobenzofuran **2a** under various conditions (Table 1). Initially, several bases Na_2CO_3, NaOH, KOH, NaO^{t}Bu, KO^{t}Bu, and NaOEt were chosen for this transformation in THF at 80 °C under nitrogen atmosphere (Table 1, entries 1–6), KOH was determined to be the best choice. To our delight, mixing the commonly used bases provided better outcomes, particularly the KOH/NaO^{t}Bu combinations (Table 1, entry 7). Further investigation of the base stoichiometry offered better product yields (Table 1, entries 8–13). Having the fruitful base combination, we next surveyed the regularly used solvents (Table 1, entries 14–19). Among the screened solvents, most promoted this transformation in low yields, making THF the best choice in terms of yield. Further variation of reaction temperatures revealed that 80 °C is the optimal condition (Table 1, entries 20, 21). Surpris-

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Table 1. Optimization of the reaction conditions.[a]



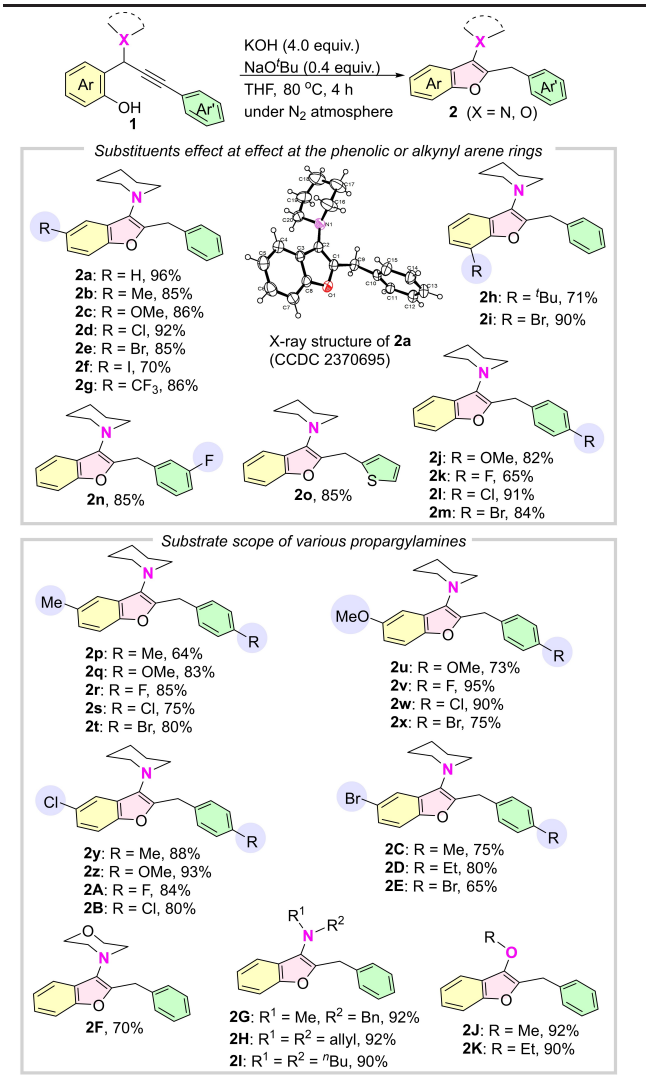
Entry	Base (equiv.)	Solvent	Temp./ °C	Time/ h	Yield/ %[b]
1	Na_2CO_3 (4.0)	THF	80	12	nr
2	NaOH (4.0)	THF	80	12	78
3	KOH (4.0)	THF	80	12	85
4	NaO^{t}Bu (4.0)	THF	80	12	80
5	KO^{t}Bu (4.0)	THF	80	12	60
6	NaOEt (4.0)	THF	80	12	40
7	KOH (4.0) /NaO^{t}Bu (1.0)	THF	80	12	92
8	KOH (3.0) /NaO^{t}Bu (1.0)	THF	80	12	80
9	KOH (5.0) /NaO^{t}Bu (1.0)	THF	80	12	85
10	KOH (4.0) /NaO^{t}Bu (0.2)	THF	80	12	86
11	KOH (4.0) /NaO^{t}Bu (0.4)	THF	80	12	92
12	KOH (4.0) /NaO^{t}Bu (0.6)	THF	80	12	90
13	KOH (4.0) /NaO^{t}Bu (1.2)	THF	80	12	88
14	KOH (4.0) /NaO^{t}Bu (0.4)	MeCN	80	12	30
15	KOH (4.0) /NaO^{t}Bu (0.4)	Acetone	80	12	35
16	KOH (4.0) /NaO^{t}Bu (0.4)	Toluene	80	12	40
17	KOH (4.0) /NaO^{t}Bu (0.4)	DME	80	12	73
18	KOH (4.0) /NaO^{t}Bu (0.4)	DMF	80	12	nr
19	KOH (4.0) /NaO^{t}Bu (0.4)	DMSO	80	12	nr
20	KOH (4.0) /NaO^{t}Bu (0.4)	THF	70	12	83
21	KOH (4.0) /NaO^{t}Bu (0.4)	THF	90	12	67
22	KOH (4.0) /NaO^{t}Bu (0.4)	THF	80	2	80
23	KOH (4.0) /NaO^{t}Bu (0.4)	THF	80	3	90
24	KOH (4.0) /NaO^{t}Bu (0.4)	THF	80	4	96
25	KOH (4.0) /NaO^{t}Bu (0.4)	THF	80	5	88

[a] Reaction conditions: 2-(3-phenyl-1-(piperidin-1-yl)prop-2-yn-1-yl)phenol **1a** (0.2 mmol) and base (as indicated) in solvent (2.0 mL) under N_2 atmosphere.
[b] Isolated yields were reported.

ingly, decreasing the reaction to 4 h, the model reaction proceeded smoothly and gave an improved yield of 96% (Table 1, entries 22–25). Therefore, the optimized conditions were eventually found as KOH/NaO^{t}Bu as the base combinations in THF at 80 °C for 4 h under N_2 atmosphere (Table 1, entry 24).

The optimal reaction conditions for the synthesis of amino-functionalized benzofurans were then used to investigate the versatility of this intramolecular annulation (Table 2). Arene substituents at the phenolic and the alkynyl arene moieties were varied to study their electronic and steric influences. The identity of the substituents at either the *para*- or *ortho*-position of the phenolic moiety has essentially no impact on yield,

Table 2. Substrate scope for the synthesis of 3-aminobenzofurans.[a]



[a] Reaction conditions: propargylamines **1** (0.2 mmol), KOH (4.0 equiv.), and NaO^{t}Bu (0.4 equiv.) in undried THF (2.0 mL) at 80 °C under N_2 atmosphere for 4 h. Isolated yields were reported.

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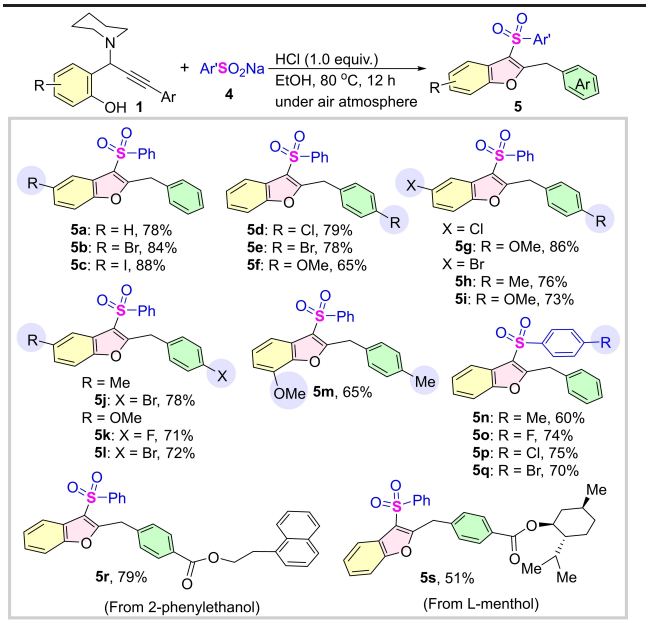
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furnishing the corresponding products **2a**–**2i** in 70–96% yields. It was noted that various functionalities, such as halo (e.g., –F, –Cl, –Br, –I), and a trifluoromethyl group, were all tolerated well, giving the desired products in decent yields, respectively. Additionally, the steric bulky *tert*-butyl group at the *ortho*-phenolic-position was found to be a feasible reaction partner toward the ring-forming process (product **2h**). Similarly, substituents including electron-rich (e.g., –OMe) group and electron-deficient (e.g., –F, –Cl, –Br) groups at the *para*- and *meta*-position of the alkynyl arene rings were also tolerable under the standard conditions. The corresponding products **2j**–**2n** were obtained in 65–91% yields. Gratifyingly, alkynyl groups bearing the 2-thienyl moiety reacted smoothly, affording the desired product **2o** in 85% yield. Subsequently, we investigated various propargylamines bearing with different substituents on both the phenolic and alkynyl arene rings to showcase the prospective utility of this protocol. Substituents with electron-rich (e.g., –Me, –OMe) groups and electron-deficient (e.g., –F, –Cl, –Br) groups at the phenolic and alkynyl arene rings were well-tolerated. The corresponding products **2p**–**2E** were obtained in good-to-excellent yields (64–95%). Meanwhile, turning to the substituents on nitrogen atom of propargylamines, when piperidinyl is replaced with morpholinyl group, the desired product **2F** was obtained in 70% yield. Moreover, changing cyclic amino group to acyclic amino groups of the substrates under the optimized conditions, the corresponding 3-aminobenzofurans **2G**, **2H**, and **2I** were obtained in 92%, 92%, and 90% yields, respectively. To our great delight, switching the amino group of the substrate **1** into the alkoxy groups resulted in the formation of the 3-alkoxyl functionalized benzofurans **2J** and **2K** in 92% and 90% yields, respectively.

To ascertain the application of the propargylamines in the construction of 3-functionalized benzofurans, we then tried introducing sodium benzenesulfinate as the substrate to form the 3-sulfonylbenzofurans *via* a cascade intramolecular annulation and C–S bond formation process. The further reaction conditions optimal showed that the desired product **5a** was successfully obtained from propargylamine **1a** and sodium benzenesulfinate **4a**, but the presence of hydrochloric acid as Brønsted acid was essential and the reaction totally failed in its absence (for details, see Supporting Information). We then investigated the substrate scope for testing the compatibility of the reaction (Table 3). The propargylamines bearing electron-donating (e.g., –Me, –OMe) and electron-withdrawing (e.g., –Cl, –Br, –I) groups could take part in this cascade reaction, delivering the corresponding products **5b**–**5m** in synthetically useful yields. It is worth noting that the halo-substituents at either the phenolic or alkynyl arene moieties were well compatible, which allows these products to be further modified by well-established transition-metal-catalyzed cross-coupling strategies at a later-stage. In addition, substituents including electron-neutral (e.g., –Me) and electron-deficient (e.g., –Cl, –Br) groups at the *para*-position of the sodium arylsulfinates **4** were well-tolerated in this transformation, delivering the corresponding products **5n**–**5q** were obtained in 60–75% yields. We further illustrated the synthetic utility of this strategy for the synthesis of 3-functionalized benzofurans containing bioactive scaffold. Interestingly, 2-phenylethanol and L-menthol derived propargylamines were proven to be viable substrates as well, affording the corresponding products **5r** and **5s** in 79% and 51% yields, respectively.

Recently, C–P bond formation reactions have attracted much attention as appealing alternatives to construct phosphorylated heterocycles.[22] After having the above successful results, we also wanted to find a novel nucleophile with P-hetero atom for the synthesis of 3-phosphinoylbenzofurans from propargylamines **1** as substrate. We delightedly noticed that commercially available diphenylphosphine oxide could also useful to form the target product **6a** in the presence of ZnI_2 as Lewis acid. After condition optimizations (see Supporting Information for details), the corresponding products **6** were obtained in satisfactory yields (Table 4). Importantly, this catalytic system also ex-

Table 3. Substrate scope for the synthesis of 3-sulfonylbenzofurans.[a]

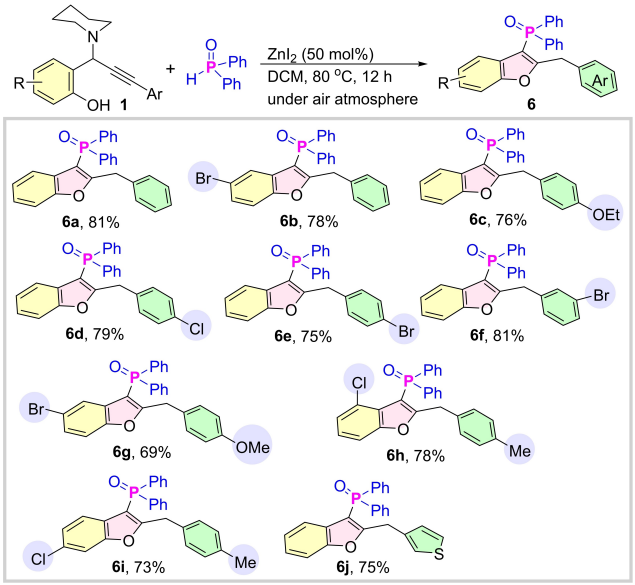


[a] Reaction conditions: propargylamines **1** (0.2 mmol), sodium arylsulfinates **4** (0.2 mmol), and hydrochloric acid (1.0 equiv.) in undried EtOH (2.0 mL) at 80 °C under air atmosphere for 12 h. Isolated yields were reported.

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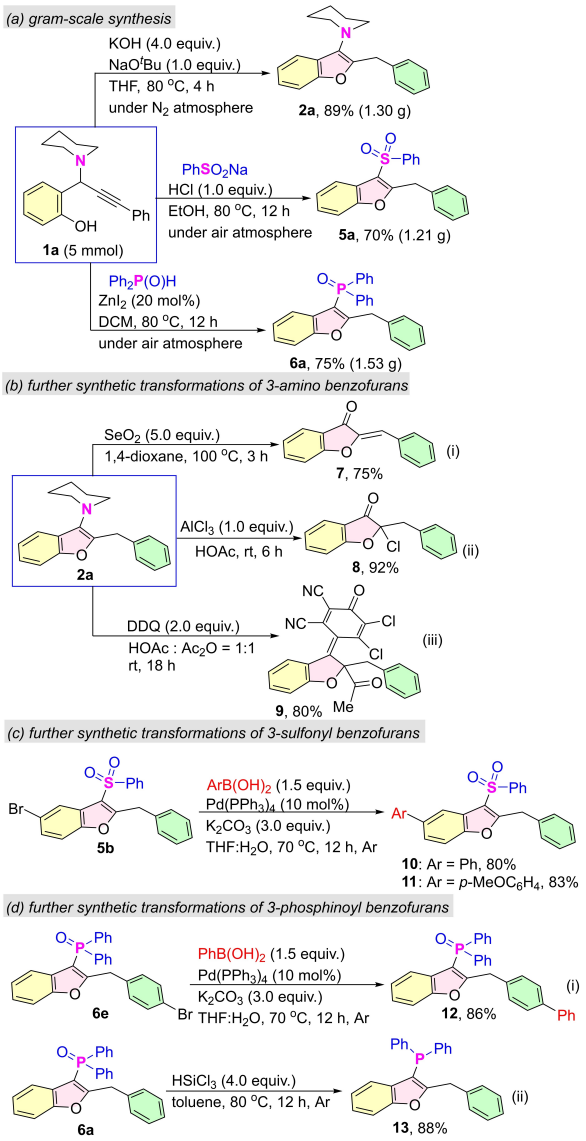
Table 4. Substrate scope for the synthesis of 3-phosphinoyl-benzofurans.[a]



[a] Reaction conditions: propargylamines **1** (0.2 mmol), diphenylphosphine oxide (0.6 mmol), and ZnI_2 (50 mol%) in undried DCM (2.0 mL) at 80 °C under air atmosphere for 12 h. Isolated yields were reported.

hibited good functional groups compatibility. In contrast with the reactions performed by Du,[15] this synthetic strategy offers remarkable operational simplicity as it allows the use of benchtop-grade solvents without the need for predrying measures to form two new C-hetroatom bonds and furan ring in a single step.

To further investigate the potential application of the above synthetic protocol, a gram-scale experiment of the 3-functionalized benzofurans was carried out. Under the standard conditions, the desired products **2 a**, **5 a**, and **6 a** were afforded successfully on a gram-scale with a comparable yield, respectively (Scheme 3a). To showcase the synthetic utilities, product **2 a** was used for further synthetic transformation. First, the reaction of **2 a** with SeO_2 delivered a 75% yield of the privileged (*Z*)-2-benzylidenebenzofuran-3(2*H*)-one **7** (Scheme 3b(i)), which molecules with such a skeleton possess potential pharmacological properties. On treatment of **2 a** with $AlCl_3$, the chloro-substituted benzofuran-3(2*H*)-one **8** was obtained in 92% yield (Scheme 3b(ii)). The product structure of **8** was unambiguously characterized by single crystal X-ray crystallography (CCDC 2370696). When treating **2 a** with DDQ (2,3-dichloro-5,6-dicyano-1,4-benzoquinone) in the mixture solvent of HOAc/Ac_2O at room temperature for 18 h, compound **9** was generated in 80% yield (Scheme 3b(iii)). The reactivity of Pd-catalyzed Suzuki-Miyaura reaction of the 3-sulfonyl and 3-phosphinoyl benzofurans was then investigated,



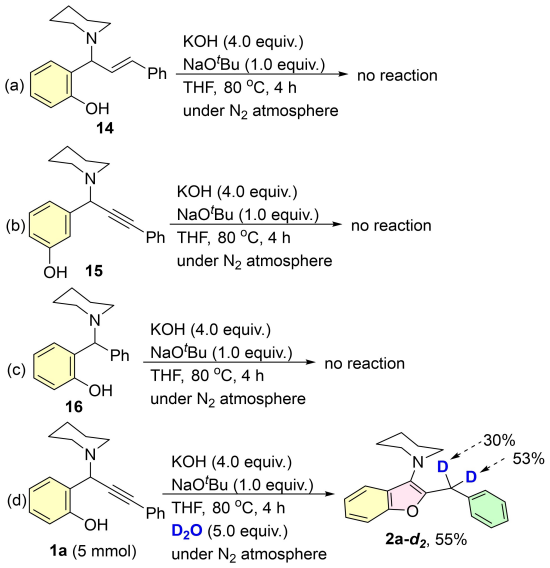
Scheme 3. Gram-scale synthesis and further synthetic transformations.

affording the corresponding products **10**, **11**, and **12** in 80%, 83%, and 86% yields, respectively (Scheme 3c and d(i)). Furthermore, the treatment of compound **6 a** with trichlorosilane also proceeded smoothly, resulting in the formation of the corresponding 3-diphenylphosphinobenfuran **13** (Scheme 3d(ii)). Accordingly, the synthetic utility of the transformation of propargylamines in the synthesis of 3-functionalized benzofurans was demonstrated.

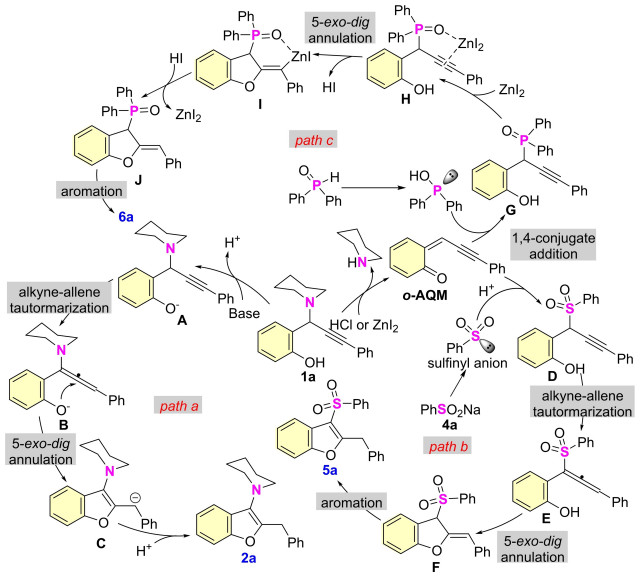
To better understand the reaction mechanism, several control experiments were performed (Scheme 3). First, (*E*)-2-(3-phenyl-1-(piperidin-1-yl)allyl) phenol **14** was synthesized and subjected to our standard conditions, but no desired product **2 a** was obtained (Scheme 4a). Similarly, no desired product **2 a** was observed by altering the hydroxyl group from

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Scheme 4. Control experiments.



Scheme 5. Proposed mechanism.

the *ortho*- to the *meta*-position on the propargylamine **15** (Scheme 4b). Furthermore, no reaction occurred when the reaction of propargylamines without the alkynyl (**16**) was performed under the standard conditions (Scheme 4c). In addition, a deuterium experiment with addition of deuterium oxide to the reaction system was carried out, and the result indicate 25% and 51% of product **2 a** was deuterated on the methylene group, respectively (Scheme 4d). These results demonstrated that the alkynyl and the *ortho*-OH group are crucial to form the furan ring through the intramolecular 5-*exo*-*dig* annulation process.

Based on the above experiment results and previous literature reports,[17c,23,24] a plausible mechanism for the formation of 3-functionalized benzofuran **2**, **5**, and **6** was proposed, respectively (Scheme 5). The mechanism for the formation of 3-aminobenzofuran **2 a** starts with the base-mediated generation of intermediate **A**, followed by the alkyne-allene tautomerization to generate the intermediate **B**. Subsequently, the oxygen anion attacks the allene group *via* 5-*exo*-*dig* annulation leading to the intermediate **C**, which then converts to the product **2 a** by trapping the proton (path a). As our previous report, the reactive *o*-AQM (detected by GC-MS, for detail see the Supporting Information) could be generated in situ from propargylamine **1 a** in the presence of Brønsted acid (HCl) or Lewis acid (ZnI_2). In path b, tautomerization of benzenesulfinate **4 a** delivers the sulfinyl anion, followed by an intermolecular 1,4-addition of *o*-AQM to afford the intermediate **D**. Then, product **5 a** is formed by the alkyne-allene tautomerization and subsequent intramolecular 5-*exo*-*dig* annulation and aromatization process. In path c, intermediate **G** is generated from the intermolecular phospha-Michael addition of nucleophilic phosphinous acid tautomerized from diphenylphosphine oxide. Next, a similar sequence of intramolecular 5-*exo*-*dig* annulation and aromatization process of the intermediate **G** furnishes the product **6 a** in the presence of ZnI_2 as Lewis acid (path c).

Conclusion

In conclusion, we have developed a facile and practical one-pot approach to useful 3-functionalized benzofuran derivatives starting from *o*-hydroxyphenyl propargylamines. A series of 3-aminobenzofurans were obtained in up to 96% yields under KOH/NaO^{t}Bu system *via* a cascade alkyne-allene tautomerization and 5-*exo*-*dig* annulation process. Additionally, commercially available benzenesulfinates or diphenylphosphine oxide were employed as nucleophile to react with the in situ formed active *o*-AQM in the presence of hydrochloric acid and ZnI_2, thus directly constructed valuable 3-sulfonyl and 3-phosphinoyl benzofuran derivatives through a sequence of 5-*exo*-*dig* annulation followed by aromatization process. Furthermore, one-pot synthesis, scale-up, and late-stage transformation of the target product highlighted the developed synthetic approaches efficacy.

Experimental Section

General Procedure for the Synthesis of 3-Amino/Alkoxy-Benzofurans 2

Under nitrogen protection, a mixture of propargylamines **1** (0.2 mmol, 1 equiv.), KOH (0.8 mmol, 4 equiv.), NaO^{t}Bu (0.08 mmol, 0.4 equiv.) and THF (2 mL) were added to a resealable screw-capped Schlenk tube. The tube sealed with a

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Teflon-coated cap and the resulting mixture was stirred in an oil bath preheated to 80 °C for 4 h (monitored by TLC). Upon completion of the reaction, the reaction mixture was cooled to room temperature, extracted with CH_2Cl_2 (3×10 mL), and washed with brine. The organic layers were combined, dried over Na_2SO_4, filtered, and then evaporated under a vacuum. The residue was purified using flash column chromatography with a silica gel (200–300 mesh), using ethyl acetate and petroleum ether (1:30, v/v) as the elution solvent to give desired products **2**.

General Procedure for the Synthesis of 3-Sulfonyl Benzofurans 5

A mixture of propargylamines **1** (0.2 mmol, 1.0 equiv.), sodium benzenesulfinate **4** (0.3 mmol, 1.5 equiv.), and HCl (1.0 equiv.) were added under air atmosphere to a resealable screw-capped Schlenk tube. EtOH (2 mL) was then added. The tube sealed with a Teflon-coated cap and the resulting mixture was stirred in an oil bath preheated to 80 °C for 12 h (monitored by TLC). Upon completion of the reaction, the reaction mixture was cooled to room temperature, extracted with CH_2Cl_2 (3×10 mL), and washed with brine. The organic layers were combined, dried over Na_2SO_4, filtered, and then evaporated under a vacuum. The residue was purified using flash column chromatography with a silica gel (200–300 mesh), using ethyl acetate and petroleum ether (1:10, v/v) as the elution solvent to give desired products **5**.

General Procedure for the Synthesis of 3-Phosphinoylbenzofurans 6

A mixture of propargylamines **1** (0.2 mmol, 1 equiv.), diphenylphosphine oxide (0.6 mmol, 3.0 equiv.), and ZnI_2 (0.1 mmol, 50 mol%) were added under air atmosphere to a resealable screw-capped Schlenk tube. DCM (2 mL) was then added. The tube sealed with a Teflon-coated cap and the resulting mixture was stirred in an oil bath preheated to 80 °C for 12 h (monitored by TLC). Upon completion of the reaction, the reaction mixture was cooled to room temperature, extracted with CH_2Cl_2 (3×10 mL), and washed with brine. The organic layers were combined, dried over Na_2SO_4, filtered, and then evaporated under a vacuum. The residue was purified using flash column chromatography with a silica gel (200–300 mesh), using ethyl acetate and petroleum ether (1:3, v/v) as the elution solvent to give desired products **6**.

Procedure for the Synthesis of Compound 7

A mixture of 1-(2-benzylbenzofuran-3-yl) piperidine **2a** (0.2 mmol, 1 equiv.), and SeO_2 (1 mmol, 5.0 equiv.) were added under air atmosphere to a resealable screw-capped Schlenk tube. 1,4-dioxane (2 mL) was then added. The tube sealed with a Teflon-coated cap and the resulting mixture was stirred in an oil bath preheated to 100 °C for 3 h (monitored by TLC). Upon completion of the reaction, the reaction mixture was cooled to room temperature, extracted with CH_2Cl_2 (3×10 mL), and washed with brine. The organic layers were combined, dried over Na_2SO_4, filtered, and then evaporated under a vacuum. The residue was purified using flash column chromatography with a silica gel (200–300 mesh), using ethyl acetate and petroleum ether (1:10, v/v) as the elution solvent to give desired product **7** in 75% yield.

Procedure for the Synthesis of Compound 8

A mixture of 1-(2-benzylbenzofuran-3-yl) piperidine **2a** (0.2 mmol, 1 equiv.), and $AlCl_3$ (0.2 mmol, 1.0 equiv.) were added under air atmosphere to a resealable screw-capped Schlenk tube. HOAc (2 mL) was then added. The tube sealed with a Teflon-coated cap and the resulting mixture was stirred at room temperature for 6 h to afford the corresponding pure products.monitored by TLC). Upon completion of the reaction, the reaction mixture was cooled to room temperature, extracted with CH_2Cl_2 (3×10 mL), and washed with brine. The organic layers were combined, dried over Na_2SO_4, filtered, and then evaporated under a vacuum. The residue was purified using flash column chromatography with a silica gel (200–300 mesh), using ethyl acetate and petroleum ether (1:4, v/v) as the elution solvent to give desired product **8** in 92% yield.

Procedure for the Synthesis of Compound 9

A mixture of 1-(2-benzylbenzofuran-3-yl) piperidine **2a** (0.2 mmol, 1 equiv.), and DDQ (0.4 mmol, 2.0 equiv.) were added under air atmosphere to a resealable screw-capped Schlenk tube. HOAc:Ac_2O = 1:1 (2 mL) was then added. The tube sealed with a Teflon-coated cap and the resulting mixture was stirred at room temperature for 8 h to afford the corresponding pure products. monitored by TLC). Upon completion of the reaction, the reaction mixture was cooled to room temperature, extracted with CH_2Cl_2 (3×10 mL), and washed with brine. The organic layers were combined, dried over Na_2SO_4, filtered, and then evaporated under a vacuum. The residue was purified using flash column chromatography with a silica gel (200–300 mesh), using ethyl acetate and petroleum ether (1:3, v/v) as the elution solvent to give desired product **9** in 80% yield.

Procedure for the Synthesis of Compound 10 and 11

Under argon atmosphere, compound **5b** (0.2 mmol, 1 equiv.), phenylboronic acid (0.3 mmol, 1.5 equiv.) or 4-methoxyphenylbornic acid (0.3 mmol, 1.5 equiv.), K_2CO_3 (0.6 mmol, 3.0 quiv.), $Pd(PPh_3)_4$ (0.02 mmol, 10 mol%), were successively added to a 15 mL Schlenk tube, followed by the addition of THF (1.0 mL) and H_2O (1.0 mL) as solvent, then heated to 70 °C for 12 h (monitored by TLC). Upon completion of the reaction, the reaction mixture was cooled to room temperature, extracted with CH_2Cl_2 (3×10 mL), and washed with brine. The organic layers were combined, dried over Na_2SO_4, filtered, and then evaporated under a vacuum. The residue was purified using flash column chromatography with a silica gel (200–300 mesh), using ethyl acetate and petroleum ether (1:10, v/v) as the elution solvent to give desired product **10** and **11** in 80% and 83% yields, respectively.

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Procedure for the Synthesis of Compound 12

Under argon atmosphere, compound **6e** (0.2 mmol, 1 equiv.), phenylboronic acid (0.3 mmol, 1.5 equiv.), K_2CO_3 (0.6 mmol, 3.0 equiv.), $Pd(PPh_3)_4$ (0.02 mmol, 10 mol%), were successively added to a 15 mL Schlenk tube, followed by the addition of THF (1.0 mL) and H_2O (1.0 mL) as solvent, then heated to 70 °C for 12 h (monitored by TLC). Upon completion of the reaction, the reaction mixture was cooled to room temperature, extracted with CH_2Cl_2 (3×10 mL), and washed with brine. The organic layers were combined, dried over Na_2SO_4, filtered, and then evaporated under a vacuum. The residue was purified using flash column chromatography with a silica gel (200–300 mesh), using ethyl acetate and petroleum ether (1:3, v/v) as the elution solvent to give desired product **12** in 86% yield.

Procedure for the Synthesis of Compound 13

Under argon atmosphere, compound **6a** (0.2 mmol, 1 equiv.) was placed with a magnetic stir bar in a Schlenk tube, then toluene (2 mL) and $HSiCl_3$ (0.8 mmol, 4.0 equiv.) were added by a syringe. The resulting solution was stirred at 80 °C for 12 h. Upon completion of the reaction, the reaction mixture was cooled to room temperature, quenched slowly by addition of 10% aq. NaOH (5 mL) and extracted with ethylacetate (3×10 mL). The combined organic layers were washed with brine. The organic layers were combined, dried over Na_2SO_4, filtered, and then evaporated under a vacuum. The residue was purified using flash column chromatography with a silica gel (200–300 mesh), using ethyl acetate and petroleum ether (1:30, v/v) as the elution solvent to give desired product **13** in 88% yield.

X-Ray Diffraction Studies of 2a and 8

CCDC-2370695 and 2370696 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre *via* www.ccdc.cam.ac.uk/data_request/cif.

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